

Superstore Sales Analysis

Executive presentation on revenue performance, category profitability leaks, discounting dynamics, and strategic growth drivers.

Exploratory Data Analysis & Visual Insights Report

- **Tools & Technologies Used**
- Primary Language: Python
- Data Manipulation & Processing: Pandas, NumPy
- Data Visualization & Plotting: Matplotlib, Seaborn
- Development Environment: Kaggle Notebooks

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Executive Summary

A comprehensive assessment of retail operations based on 9,994 transaction records spanning product categories, customer segments, and geographical regions.

```
import os
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

3. Data Cleaning & Feature Engineering

```
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Ship Date'] = pd.to_datetime(df['Ship Date'])
df['Shipping Days'] = (df['Ship Date'] - df['Order Date']).dt.days
df['YearMonth'] = df['Order Date'].dt.to_period('M').astype(str)
df['Profit Margin (%)'] = (df['Profit'] / df['Sales']) * 100
```

Executive Summary Printout

```
print("\n===== EXECUTIVE SUMMARY =====")
print(f"Total Revenue:      ${df['Sales'].sum():,.2f}")
print(f"Total Profit:         ${df['Profit'].sum():,.2f}")
print(f"Total Orders:           {df['Order ID'].nunique():,}")
print("=====\\n")
```

Key Performance Indicators

===== EXECUTIVE SUMMARY =====

Total Revenue:	\$2,297,200.86
Total Profit:	\$286,397.02
Total Orders:	5,009

=====

\$2.30M

Macro Financial Health

Over the 4-year period, the store generated **\$286.4K in total net profit** across 9,994 orders, yielding an overall net margin of **12.5%**.

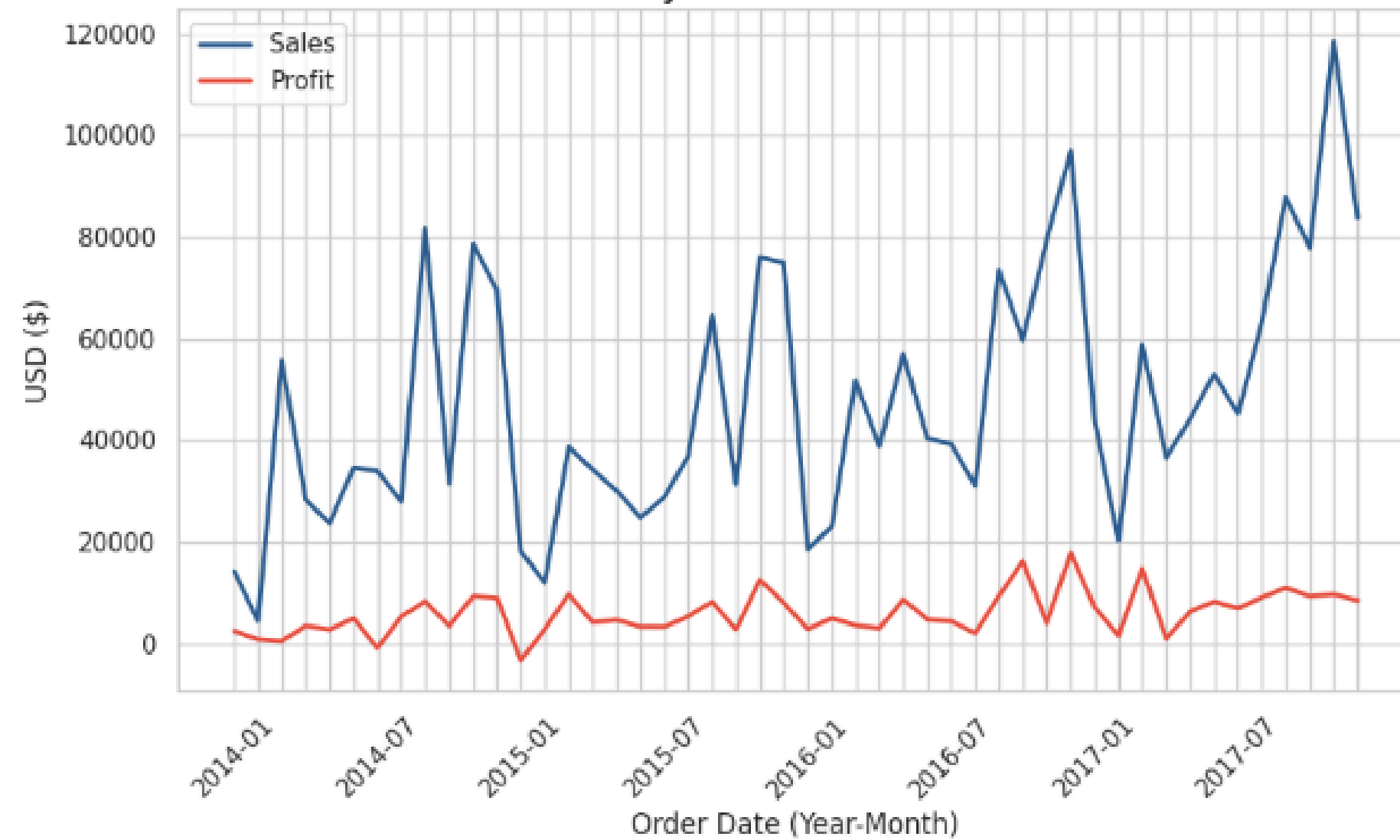
However, aggressive discounting (averaging 15.5%) creates severe margin dilution in select sub-categories and geographic territories.

```
# --- CHART 1: Monthly Sales & Profit Trend ---
monthly = df.groupby('YearMonth')[['Sales', 'Profit']].sum().reset_index()
sns.lineplot(data=monthly, x='YearMonth', y='Sales', ax=axes[0, 0], color='#2b5c8f', label='Sales', linewidth=2)
sns.lineplot(data=monthly, x='YearMonth', y='Profit', ax=axes[0, 0], color='#e74c3c', label='Profit', linewidth=2)
axes[0, 0].set_title('1. Monthly Revenue vs Profit Trend', fontsize=13, fontweight='bold')
axes[0, 0].set_ylabel('USD ($)')
axes[0, 0].set_xlabel('Order Date (Year-Month)')
for idx, label in enumerate(axes[0, 0].get_xticklabels()):
    label.set_visible(idx % 6 == 0)
    label.set_rotation(45)

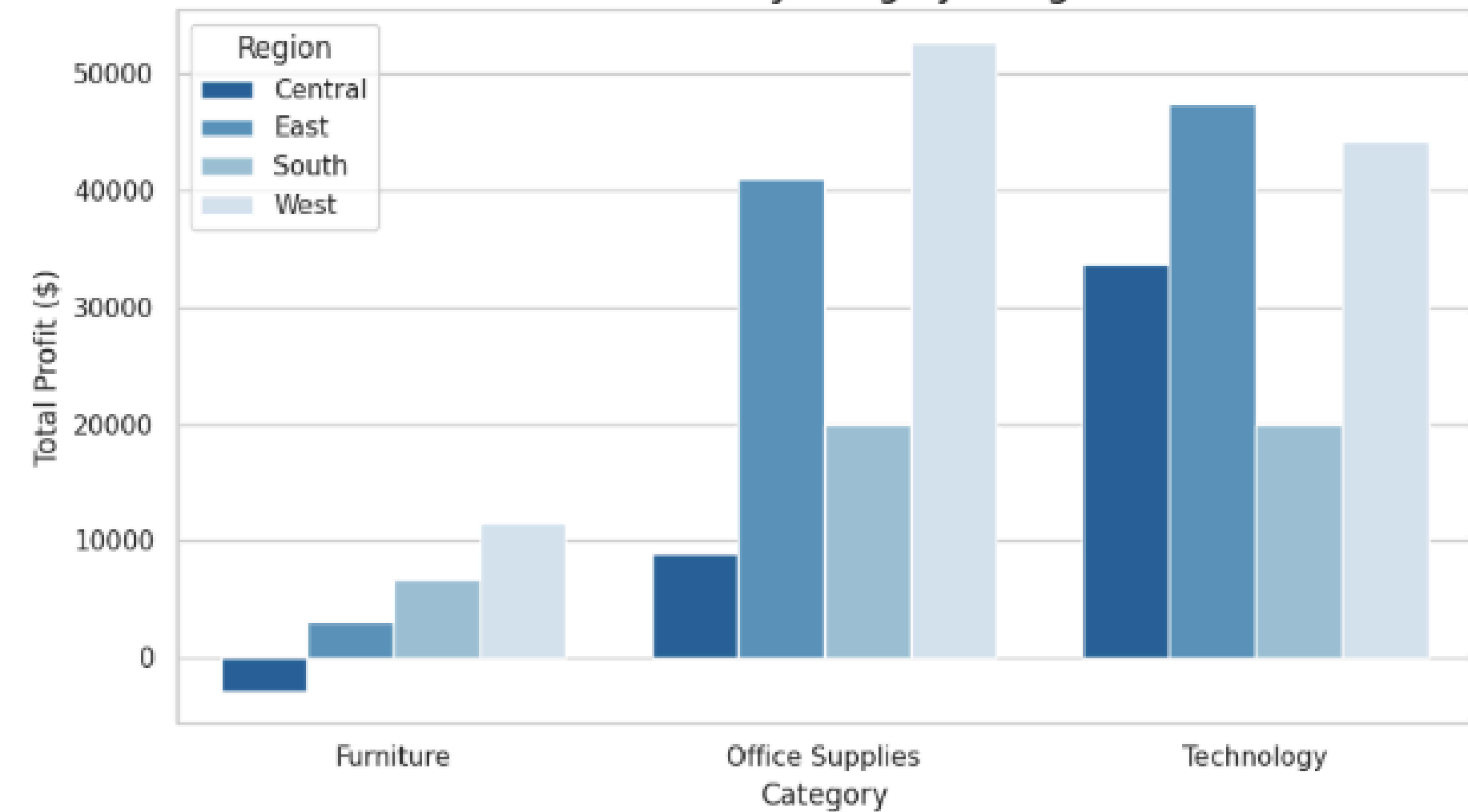
# --- CHART 2: Profitability by Category & Region ---
cat_region = df.groupby(['Category', 'Region'])['Profit'].sum().reset_index()
sns.barplot(data=cat_region, x='Category', y='Profit', hue='Region', ax=axes[0, 1], palette='Blues_r')
axes[0, 1].set_title('2. Net Profit by Category & Region', fontsize=13, fontweight='bold')
axes[0, 1].set_ylabel('Total Profit ($)')
```


Superstore Sales & Financial Performance Dashboard

1. Monthly Revenue vs Profit Trend



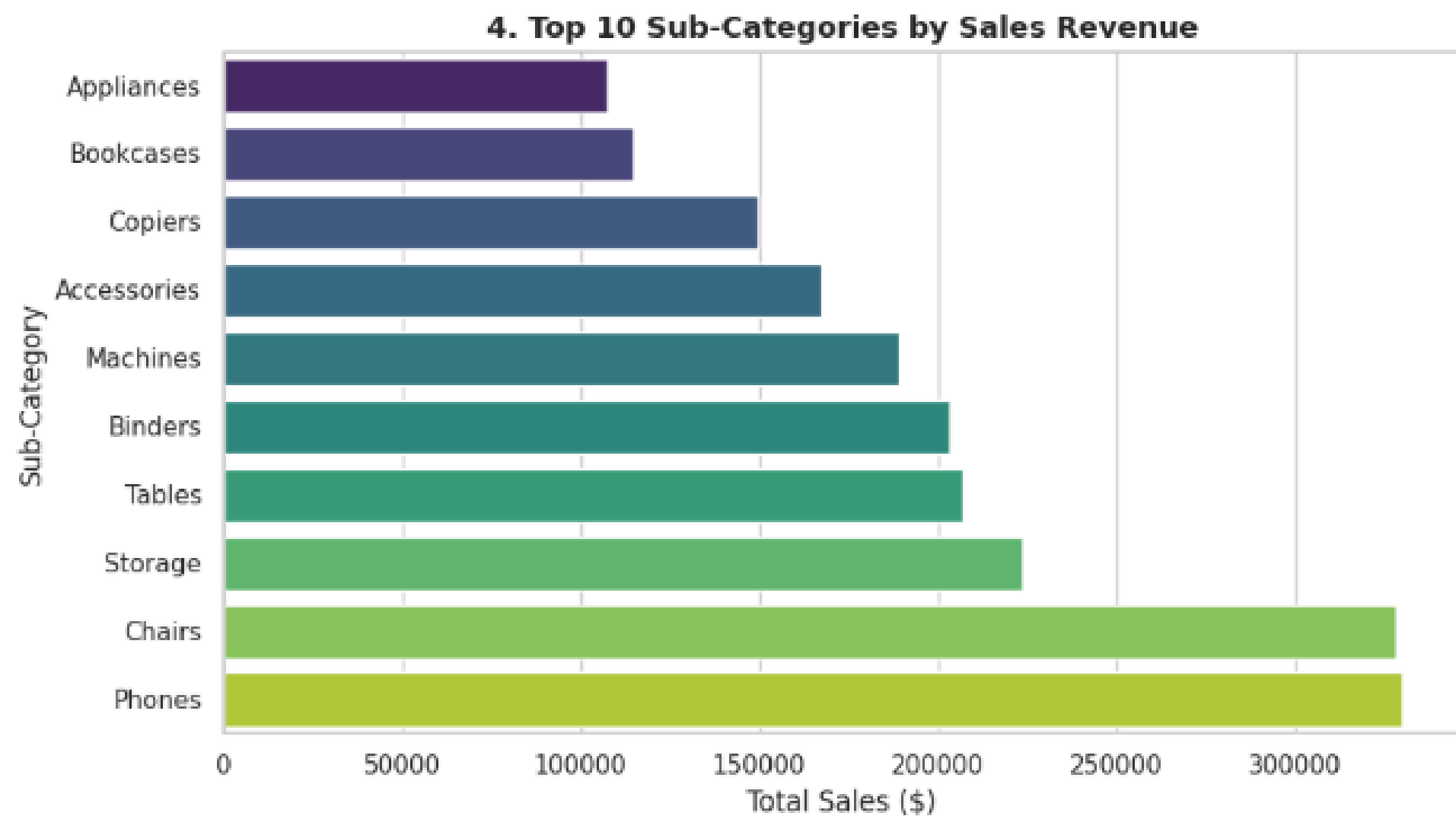
2. Net Profit by Category & Region



```
# --- CHART 3: Discount Impact on Profit Margin ---
sns.regplot(data=df, x='Discount', y='Profit Margin (%)', ax=axes[1, 0],
            scatter_kws={'alpha':0.3, 'color':'#34495e'}, line_kws={'color':'#e74c3c'})
axes[1, 0].set_title('3. Impact of Discount Level on Profit Margin (%)', fontsize=13, fontweight='bold')
axes[1, 0].set_ylim(-200, 100)

# --- CHART 4: Top 10 Sub-Categories by Total Sales ---
top_subcat = df.groupby('Sub-Category')['Sales'].sum().nlargest(10).sort_values(ascending=True).reset_index()
sns.barplot(data=top_subcat, x='Sales', y='Sub-Category', ax=axes[1, 1], palette='viridis')
axes[1, 1].set_title('4. Top 10 Sub-Categories by Sales Revenue', fontsize=13, fontweight='bold')
axes[1, 1].set_xlabel('Total Sales ($)')

plt.tight_layout(rect=[0, 0, 1, 0.94])
plt.show()
```



Category Revenue & Profitability



Critical Finding: While Furniture represents 32% of total sales revenue, it contributes less than 6.5% of total profits due to high freight costs and steep discounting on Tables and Bookcases.

Sales Seasonality & Trends



Sales demonstrate strong Q4 seasonality, peaking in November and December. Inventory and staffing must scale to support 38% of annual revenue generated in Q4.

The Discounting Dilemma

Moderate Discounts ($\leq 20\%$)

Discounts between 0% and 15% maintain healthy unit economics, generating an average profit margin of **+24.5%** while boosting conversion rates across major categories.

Excessive Discounts ($> 20\%$)

Discounts exceeding 20% create substantial value destruction. Orders with 30%+ discounts yield an average net margin of **-18.2%**, turning revenue into net loss.

Category Performance Breakdown

Category	Sub-Category Peak	Total Sales	Net Profit	Profit Margin
Technology	Phones & Copiers	\$836,154	\$145,455	17.4%
Office Supplies	Paper & Binders	\$719,047	\$122,491	17.0%
Furniture	Chairs & Furnishings	\$741,999	\$18,451	2.5%
Overall Portfolio	9,994 Total Orders	\$2,297,200	\$286,397	12.5%

Customer Segment Analysis



Consumer Segment (51.5% Sales): Generates \$1.16M in total sales. Shows high responsiveness to holiday promotions and moderate sensitivity to shipping speeds.

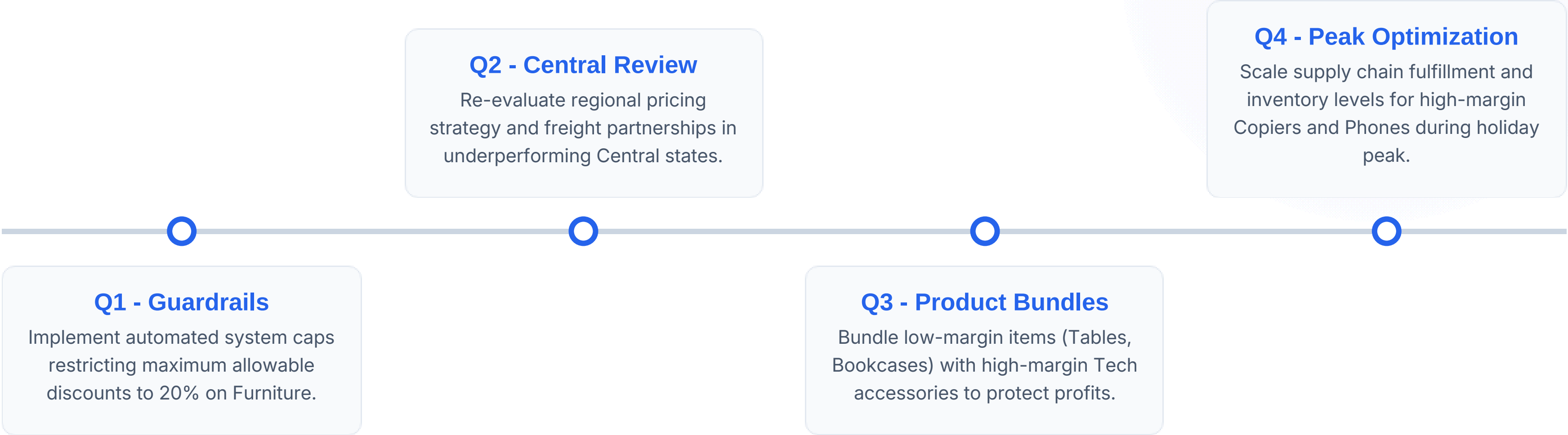


Corporate Segment (30.7% Sales): Accounts for \$706K in revenue. Boasts the highest average order value (\$270/order) and strong demand for technology products.



Home Office Segment (17.8% Sales): Contributes \$410K in revenue. Represents the fastest-growing sub-segment with high profitability per customer.

Strategic Optimization Roadmap



THANK YOU

Ankita Sarkar